

Joshua Prettyman, Ph.D.

Data Scientist, Mathematics Ph.D., Python Developer

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PERSONAL STATEMENT

I built an ML-powered SaaS platform that increased SEO productivity by 20×, taking Blink SEO's internal tooling from spreadsheets to a marketable product now used by multiple agencies: *Macaroni Software*. I created the role after being hired as a one-man data/dev team with the brief: "use data to improve processes".

My research at the National Physical Laboratory produced a novel scaling indicator for tipping point detection in multidimensional time series, published in *Environmental Research Letters*, 2022.

I am seeking a remote data science role where I can build production systems and continue solving complex technical problems.

TECHNICAL SKILLS

- Regular use of Python libraries: ScikitLearn, Numpy, Pandas, NLTK, Matplotlib, Plotly.
- Python development: Data Science/ML, async, package management, REST, GraphQL.
- Full stack development with Bash, Python, Google Cloud Platform, SQL and JavaScript.
- Huggingface transformers and LLMs, Ollama, OpenAI Assistant API.
- Experience with C++, MATLAB, Retool, data engineering, dashboards, scientific computing.

PROFESSIONAL EXPERIENCE

Data Scientist — Blink SEO / Macaroni Software

Nov. 2021 to Dec. 2024

Built a full-stack ML platform from scratch, increasing team productivity by 20× and enabling year-long SEO campaigns to be delivered in weeks.

- Developed NLP-powered keyword clustering, combined with quantitative data, reducing manual work from days to minutes.
- Architected Python backend (*GCP Compute Engine*) integrated with *BigQuery* to process 50M+ data points daily from multiple APIs.
- Worked with the team to replicate and then improve their (spreadsheet-based) processes with continual feedback and deployment.
- Created a job-queue system (*PostgreSQL*) to run client onboarding, data imports, data analysis and ML tasks asynchronously in the backend.
- Built interactive dashboards (*Plotly + JavaScript*) enabling clients to identify and prioritise revenue opportunities in minutes.
- Integrated LLMs (*huggingface*) to generate ready-to-go suggestions for content gaps, allowing customers to see immediate impact.
- Delivered ad-hoc analyses and data engineering for C-suite investor presentations and internal reporting.

Data Science Researcher — National Physical Laboratory (NPL)

Sep. 2015 to Feb. 2021

Produced a novel scaling indicator for tipping point detection in multidimensional time series.

- Presented results for stakeholder meetings at NPL, in published articles, and at international conferences.
- Developed methods to detect tipping points in dynamical systems, working with large time series datasets and building stochastic models to test hypotheses.
- Code written in *MATLAB* and *Python*. Visualisations produced using *Matplotlib* and *XMGrace*.
- Accessing, cleaning, interpolating and correctly interpreting large, raw meteorological datasets.

PROFESSIONAL EXPERIENCE (continued)

Associate Lecturer — Sheffield Hallam University

Sep. 2017 to Jul. 2019

Taught mathematics, statistics and computing courses from Foundation Degree to Masters.

- Taught a course on *Microsoft Excel for Business*.
- Responsibilities included lesson preparation, course planning, marking, leading tutorials.
- Manned the “Maths Help” desk three times a week, providing support to students of all departments.

Informatics developer (internship) — UK Met Office Informatics Lab

Jun. to Aug. 2017

A three-month project assessing the viability of using live data from amateur meteorologists to augment official observation data.

- Accessed internal and external data via APIs using Java.
 - Data cleaning and analyses in Python.
 - Worked as part of a team and participated in daily stand-ups.
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ACADEMIC BACKGROUND

Ph.D. Mathematics — University of Reading

Thesis: *Tipping Points and Early Warning Signals with Applications to Geophysical Data*.

- Thesis develops spectral analysis methods to detect correlations in multidimensional time series, which are used as Early Warning Signals for tipping events in stochastic dynamical systems.
- Publication of three papers in respected journals:
 - * *A novel scaling indicator of early warning signals* (EPL, 2018)
 - * *Generalized early warning signals in multivariate and gridded data* (Chaos, 2019)
 - * *Power spectrum scaling as a measure of critical slowing down* (ERL, 2022)
- Implemented mathematically derived methods numerically by discretising and linearising.
- Participation in several international conferences and workshops.
- Three-month data-focussed internship at the UK Met Office (see *Professional Exp.* section).

M.Res. Mathematics — Imperial College London (*Distinction*)

Dissertation: *Mesh Generation Using Solutions of an Optimal Transport Problem*.

- Dissertation uses solutions to a linearisation of the Monge-Ampère equation to generate an adaptive, optimally distributed mesh for numerical models.
- Working in C++; monitoring CPU usage and convergence time over various scenarios.
- Participation in weekly soft skills workshops and monthly internal showcases.
- Attendance at weekly Mathematics of Planet Earth seminars hosted by the Centre for Doctoral Training.
- Completed taught courses in Python, R, Dynamical Systems, Stochastic differential equations, Bayesian Analysis and Probability.

MA Mathematics — University of Edinburgh (*First Class Hons.*)

Dissertation: *Growth and its Applications in Graph Theory*.

- Dissertation explores patterns in paths through connected digraphs.
- For the dissertation project I wrote a program with a simple UI to investigate the properties of these digraphs: this then became a teaching aid, earning a letter of thanks from the principal.
- Requisite taught courses for the Pure Mathematics degree besides optional modules in computing and mathematics education.